Assignment 3

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September 2, 2025

1 Objective

Discuss feature extraction power of your favorite CNN pretrained by ImageNet datset before and after transfer learning by MNIST didgit dataset by plotting high dimensional feature vectors on 2D plane using three dimension reduction techniques.

2 Methodology

1. **Dataset Preparation:** The MNIST dataset was resized from 28×28 grayscale images to 48×48 RGB images for compatibility with VGG16. Normalization and `ImageDataGenerator` were used to handle memory efficiently.

2. **Feature Extraction (Before Transfer Learning):** VGG16 pretrained on ImageNet was used as a fixed feature extractor. Features were extracted from the Global Average Pooling layer.

3. **Transfer Learning:** The convolutional base of VGG16 was frozen. Fully connected layers were added and fine-tuned on MNIST. The model was trained for 5 epochs using the Adam optimizer.

4. **Feature Extraction (After Transfer Learning):** Features were again extracted from the penultimate dense layer after training.

5. **Dimensionality Reduction:**
   - PCA (Principal Component Analysis): Linear method for variance-based visualization.
   - t-SNE (t-distributed Stochastic Neighbor Embedding): Non-linear method for preserving local neighborhoods.
   - LDA (Linear Discriminant Analysis): Supervised method maximizing between-class variance.

6. **Visualization:** Feature vectors were projected into 2D space using PCA, t-SNE, and LDA. Side-by-side comparisons of before and after transfer learning were generated.

3 Result

The following figures show feature distribution before and after transfer learning:

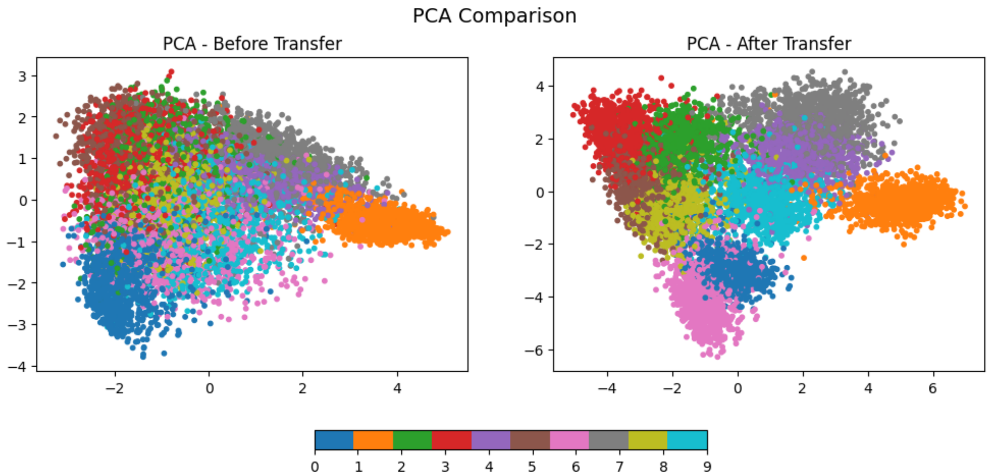

Figure 1: PCA comparison of features before and after transfer learning.

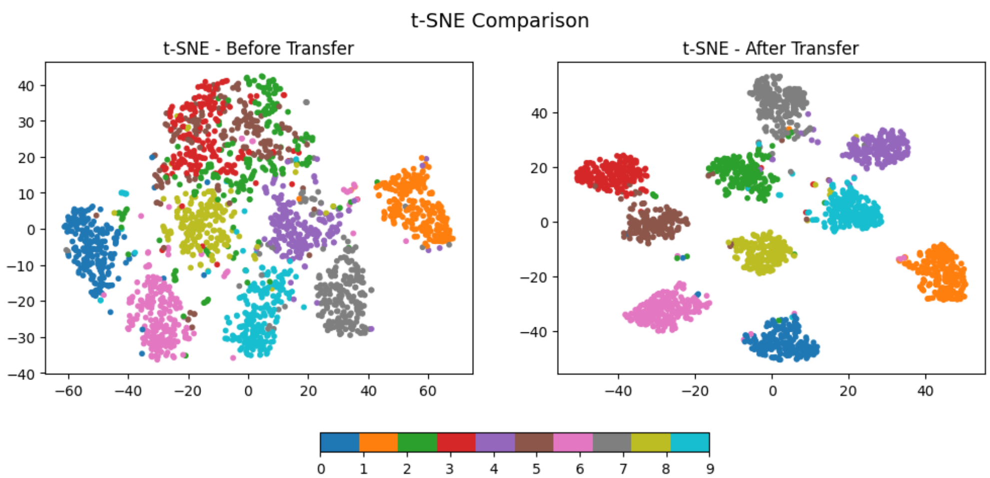

Figure 2: t-SNE comparison of features before and after transfer learning.

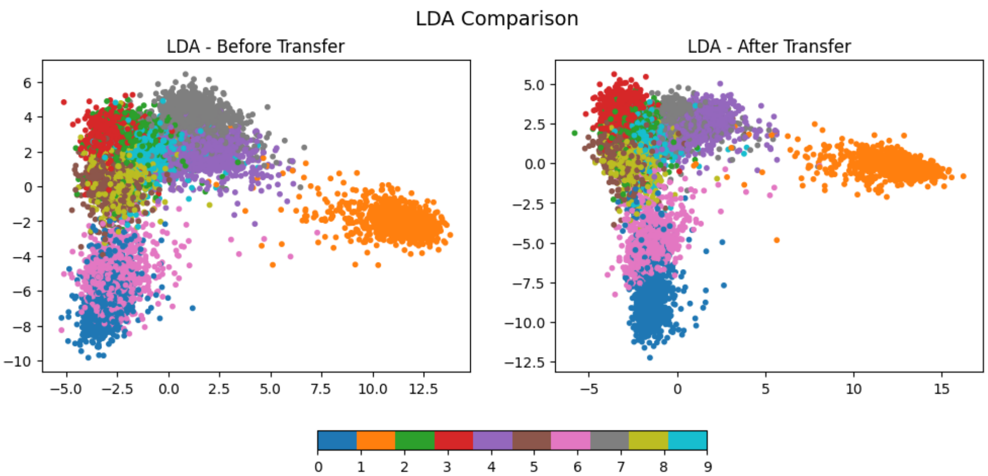

Figure 3: LDA comparison of features before and after transfer learning.

4 Analysis

- **PCA:** Before transfer, the features were messy and overlapping, so digits looked similar in feature space. After transfer, the clusters became clearer, meaning the model learned to separate digits better.
- **t-SNE:** Before transfer, the points were scattered and mixed, so digits weren't grouped well. After transfer, each digit formed a tight group with clear boundaries.
- **LDA:** Before transfer, classes were overlapping a lot, showing the features didn't capture digit identity well. After transfer, the digits were clearly separated, proving the fine-tuned model aligned features with the actual digit classes.

Source: Github